SoccerGMR Ablation Study: DQ-CGP & SI-Bind Inter-layer Binding (9 Backbones)

Comparing GMR Baseline vs. DQ-CGPv3 vs. GEQ Selection vs. SI-Bind (Inter-layer Residual Binding & Dominance Refresh) on Standard Test (1,036 queries) and Full-Decodable Test (2,970 queries).

		Standard Test Split (1,036 queries)				Full-Decodable Test Split (2,970 queries)			
Backbone	**Configuration**	**mAP (%)**	**Orc-Gap (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**	**mAP (%)**	**Orc-Gap (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**
Moment-DETR	1. GMR Baseline	6.68	24.28	67.94	70.20	5.47	23.48	72.08	78.91
	2. GMR + DQ-CGPv3	10.66	22.49	72.20	72.09	9.99	21.83	74.92	80.24
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	12.64	**15.62**	**75.48**	73.48	11.63	**15.40**	**78.93**	80.37
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	12.63	23.44	68.77	**75.17**	11.56	22.53	69.83	**81.94**
	5. GMR + SI-Bind (Ours)	**15.21**	23.66	65.52	74.32	**12.58**	22.21	68.83	81.86
EaTR	1. GMR Baseline	16.49	21.51	65.60	76.56	15.19	20.39	67.38	81.59
	2. GMR + DQ-CGPv3	16.02	23.87	65.27	**77.19**	15.35	22.31	65.29	80.38
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	17.48	**16.31**	**75.43**	76.17	15.67	**15.48**	**78.73**	**83.04**
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	**17.49**	22.74	70.12	76.28	**15.98**	22.10	72.31	82.28
	5. GMR + SI-Bind (Ours)	17.12	21.80	67.74	76.23	15.35	20.89	68.66	81.03
QD-DETR	1. GMR Baseline	14.62	20.23	71.48	**77.01**	12.65	19.05	72.49	81.75
	2. GMR + DQ-CGPv3	15.82	20.47	69.02	74.24	13.93	19.23	71.63	81.64
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	**17.28**	**18.19**	72.25	75.18	**14.20**	**17.07**	75.61	**82.48**
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	16.80	19.15	68.29	74.73	13.61	17.68	71.15	82.20
	5. GMR + SI-Bind (Ours)	16.30	20.65	**73.71**	76.63	13.20	19.84	**75.91**	82.47
CG-DETR	1. GMR Baseline	**18.65**	25.66	57.47	**76.40**	**14.67**	23.28	63.23	81.81
	2. GMR + DQ-CGPv3	13.49	22.05	75.04	74.63	12.49	20.73	76.97	79.33
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	14.75	**15.16**	75.31	75.77	13.17	**14.53**	**78.87**	**82.88**
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	14.75	25.36	58.42	75.41	13.62	23.73	62.98	82.44
	5. GMR + SI-Bind (Ours)	13.80	21.77	**75.72**	75.42	13.96	21.14	77.43	81.33
BM-DETR	1. GMR Baseline	4.00	17.59	**75.00**	71.55	3.65	17.62	78.58	79.11
	2. GMR + DQ-CGPv3	7.74	17.40	74.43	70.69	7.32	16.94	77.84	79.64
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	7.82	**12.32**	**75.00**	72.09	7.32	**12.38**	**78.80**	79.15
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	**8.69**	21.58	71.33	**73.71**	**7.82**	20.87	75.15	**80.15**
	5. GMR + SI-Bind (Ours)	7.72	20.91	73.40	70.18	7.03	20.30	77.47	79.27
BAM-DETR	1. GMR Baseline	23.25	19.82	70.43	**78.66**	20.62	18.37	71.79	80.99
	2. GMR + DQ-CGPv3	21.52	19.40	71.83	77.80	19.57	18.04	72.01	80.13
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	22.84	**17.60**	**74.07**	77.41	20.74	**16.77**	**77.58**	**83.32**
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	22.83	21.91	65.46	77.08	21.21	20.99	68.07	82.87
	5. GMR + SI-Bind (Ours)	**23.78**	20.40	68.96	78.37	**21.78**	19.07	69.28	78.69
Sim-DETR	1. GMR Baseline	20.73	22.51	74.69	72.97	19.18	21.78	75.72	78.87
	2. GMR + DQ-CGPv3	20.63	21.16	75.43	75.84	19.62	20.57	75.69	80.20
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	21.43	**15.44**	**75.56**	**76.79**	19.50	**14.77**	76.12	**82.41**
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	**21.71**	24.40	67.73	76.70	**20.19**	22.89	70.09	80.95
	5. GMR + SI-Bind (Ours)	7.27	19.90	75.00	69.30	6.90	19.42	**77.92**	78.23
UVCOM	1. GMR Baseline	7.12	**17.24**	**75.00**	70.03	6.40	16.83	**76.80**	**78.98**
	2. GMR + DQ-CGPv3	17.06	21.36	70.06	76.83	16.41	20.78	67.20	75.32
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	18.40	18.48	72.28	76.65	17.12	17.66	71.45	76.29
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	17.16	23.49	59.80	75.83	16.50	22.37	59.55	75.48
	5. GMR + SI-Bind (Ours)	**20.49**	17.55	69.48	**78.13**	**19.76**	**16.74**	68.12	75.69
FlashVTG	1. GMR Baseline	25.43	25.47	66.35	74.30	20.43	24.13	68.48	82.21
	2. GMR + DQ-CGPv3	**26.13**	22.36	67.25	**76.02**	**21.11**	20.90	68.95	**82.76**
	3. GMR + DQ-CGPv3 + GEQ (G1 Sel)	25.81	**17.57**	**74.62**	74.69	20.87	**16.52**	**78.13**	81.95
	4. GMR + DQ-CGPv3 + GEQ (mAP Sel)	26.11	20.69	73.23	74.34	21.07	19.75	76.92	81.91
	5. GMR + SI-Bind (Ours)	*-- running / pending --*				*-- running / pending --*			

Method 3 (G1 Sel): Global Event Query fine-tuned with model selection on Val Mean G-mIoU@1 (0.4/0.5/0.6) (Official).
Method 4 (mAP Sel): Global Event Query fine-tuned with model selection on highest Val Localization mAP.
Method 5 (SI-Bind): Inter-layer Residual Binding with Dominance-Aware Slot Refresh and Slot-Private Innovation (Ours).
Orc-Gap (%): Area between Oracle envelope and G-mIoU@1 threshold curve across tau in [0, 1] (Integral of Oracle_G1 minus G1-AUC).
Note: Lower Orc-Gap is better (bold marks lowest gap per backbone), directly quantifying G-mIoU loss caused by imperfect query existence gating.